Data Analyst

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Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

I'll start by examining the dataset to understand its structure, key characteristics, and contents. Then, you can provide a specific analytical question for me to address. Let me first inspect the file.

Dataset Overview:

- **Total Rows:** 221,626
- **Total Columns:** 19
- **Key Columns:**
  - `StateCode`, `MunicipalityCode` (Geographic identifiers)
  - `MonthIssued` (Contract issuance month)
  - `ProgramCode`, `SubProgramCode`, `ResourceSourceCode` (Categorical codes)
  - `FundingValue`, `InvestmentValue`, `CommercializationValue`, `IndustrializationValue` (Monetary values)
  - `ContractTotalValue` (Total contract amount)

Now, what specific analytical question would you like me to answer using this dataset? [>-]

Which attribute represents the greatest information gain of the entire dataset?

To determine which attribute provides the greatest **information gain**, I will use **entropy-based feature selection**. This involves:

1. **Calculating entropy** for the target variable (`ContractTotalValue`).
2. **Computing information gain** for each feature by measuring how much it reduces entropy.
3. **Identifying the attribute with the highest information gain** as the most informative.

I'll proceed with this analysis now.

The attribute `FundingValue` provides the greatest information gain in the dataset, with a **mutual information score of 4.779**.

This means `FundingValue` is the most informative feature for predicting `ContractTotalValue`. Let me know if you need further insights! [>-]

Message Data Analyst